

EXPLORING OMIC RESPONSES TO EXERCISE

Decoding Sexually Dimorphic Metabolomic Signatures of Exercise



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Background

Our previous transcriptomic analyses of human skeletal muscle identified exercise responsive signatures. To explore systemic cross-talk and the influence of biological sex, we utilized the NIH MoTrPAC Data Hub to investigate plasma metabolomics. We examined the association between resting plasma metabolic profiles (48h post-exercise) and cardiorespiratory fitness improvements ($\Delta\text{VO}_2\text{ max}$) following 4 and 8 weeks of endurance training.

Methodology

Dataset: MoTrPAC F344 rats (n=5/sex/group) at 4-week and 8-week time points.

Target Metabolites: Filtered for training-regulated plasma metabolites (MoTrPAC primary findings) to ensure biological relevance.

Outcome Definition: Exercise adaptation was quantified as $\%\Delta\text{VO}_2\text{ max}$ (percentage change from baseline).

Statistical Modeling: To maximize power within small-sample constraints, we utilized linear interaction models (N = 10 per timepoint): $\%\Delta\text{VO}_2 \sim \text{Metabolite_Intensity} * \text{Sex}$

Validation: Benjamini-Hochberg FDR correction was applied to differentiate between universal (Main) and sex-dimorphic (Interaction) associations.

Result 1: Week 4 - The Universal "Metabolic Stress"

Finding: Early fitness gains are characterized by sex-agnostic metabolic responses.

Key Markers: Amino acid and TCA cycle intermediates (e.g., Sarcosine, Aspartic acid) showed consistent main effects (See Figure 1).

Mechanism: Both sexes appear to prioritize a universal upregulation of energy conversion to cope with initial physiological stress. Notably, no sex-divergent adaptations were detected at this early stage.

Result 2: Week 8 - Divergent Metabolic Trajectories

Finding: Chronic adaptation transitions into specialized, sex-dependent metabolic remodeling.

Key Markers: The initial universal markers subsided, giving way to strong Sex-by-Metabolite interactions (See Figure 2).

1. Females: Positive correlations with stress regulators and lysophospholipids
2. Males: Inverse correlations with these exact same markers.

Mechanism: The divergent profiles suggest females adapt via the active upregulation of neuro-immune modulation and antioxidant defense. Conversely, the inverse trend in males suggests adaptation occurs through the enhanced clearance of these stress-related metabolites.

Objectives

To explore potential correlations between circulating plasma metabolites and improvements in fitness.

To examine whether these metabolic associations differ across biological sex and training duration.

Figure 1: Universal Energy Adaptation at Week 4

Top biomarkers exhibit main effects. A single trend line illustrates the shared adaptation.

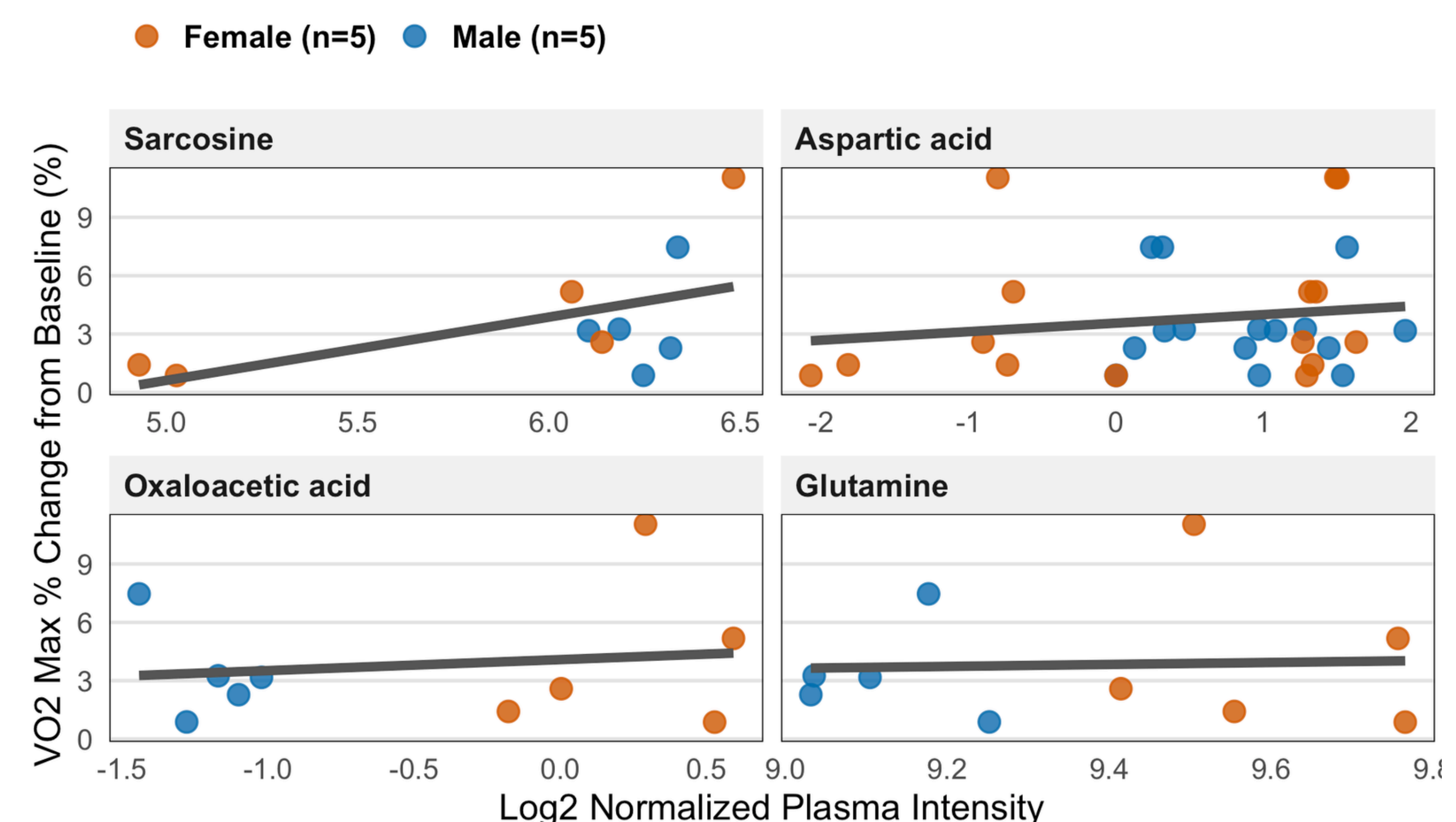


Figure 1: Universal adaptation at Week 4. Parallel trend lines demonstrate a shared upregulation of early energy metabolism across both sexes.

Figure 2: Sexually Dimorphic Exercise Adaptation at Week 8

Top biomarkers exhibit strong Sex $\times$ Metabolite interactions.

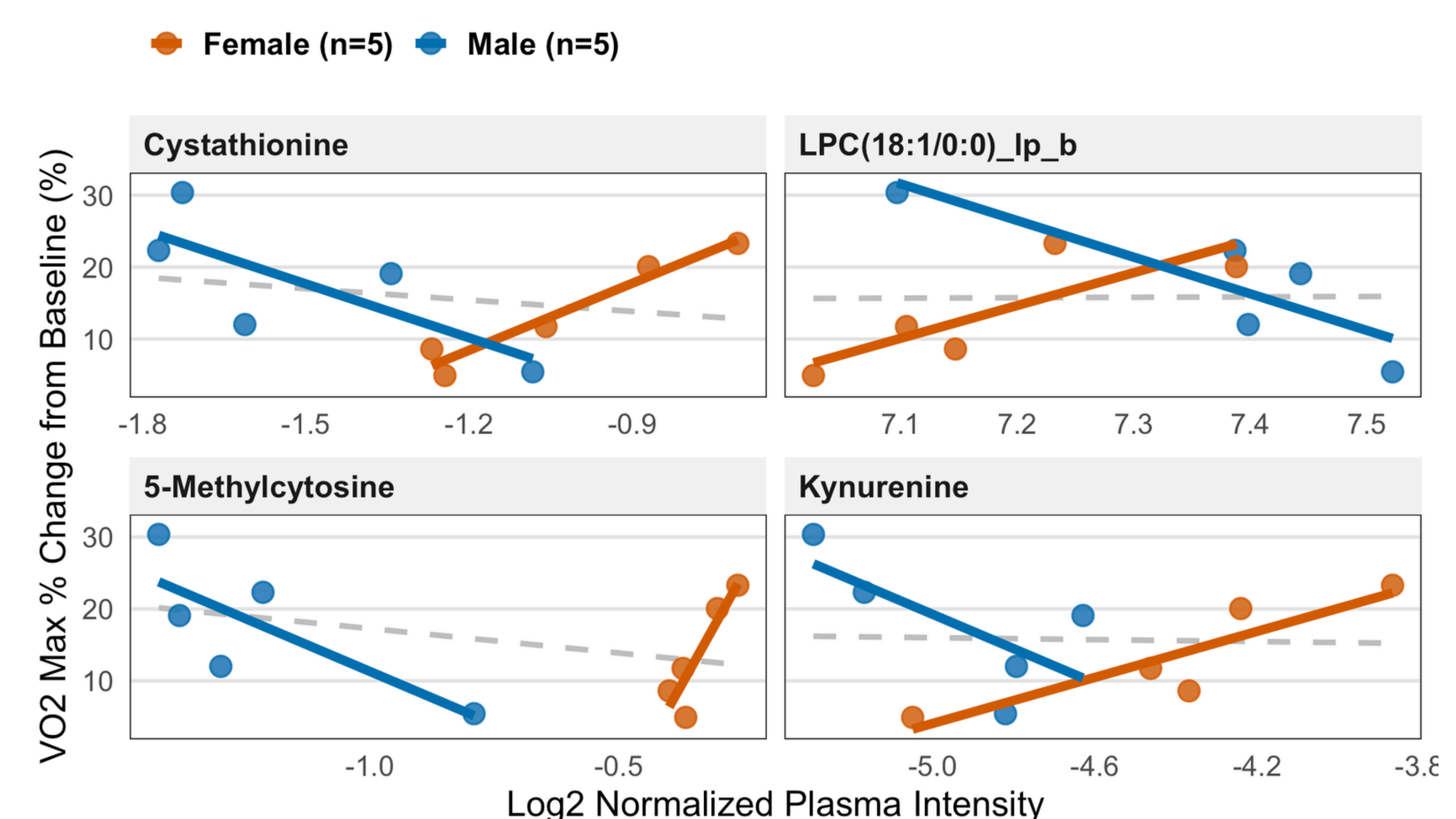


Figure 2: Sexually dimorphic adaptation at Week 8. Divergent trend lines reveal highly significant Sex $\times$ Metabolite interactions, proving that chronic adaptations are fundamentally dimorphic.

Conclusion

A Dynamic Sequence: Exercise adaptation transitions from an early, universal response to sex-specific metabolic strategies.

Precision Exercise Physiology: These exploratory findings suggest that biological sex may be a key variable in long-term exercise adaptation, highlighting the potential limitations of sex-agnostic biomarkers.

References

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Acknowledgements

I would like to express my gratitude to Prof. Andrey Skripnikov for his valuable statistical insights, and to Prof. Bernhard Klingenberg for his ongoing academic guidance and support. Data used in the preparation of this project were obtained from the Molecular Transducers of Physical Activity Consortium (MoTrPAC) database (motrpac-data.org). I also acknowledge the use of the GSE47881 dataset from the Gene Expression Omnibus.